Supplementary Information: Experiment-Driven Bayesian Optimization of the Impact Response of Multi-Material 3D-Printed Tensegrity-Inspired Structures

Draft note (remove before submission): as in the main text, any value printed in this violet color is a synthetic placeholder rather than a measurement. As of 2026-09-15 this Supplementary Information contains no violet values. The former placeholder batch-2 print key was replaced by the logged campaign record, which now covers batches 2 to 4.

This Supplementary Information collects the detailed design-, fabrication-, test-, and analysis-development records supporting the main manuscript, together with the tables and figures moved out of the main text to meet the journal's length limit (the main text cites them by their S-numbers). Unlike the main text, the SI references the project's development history directly, including specific pull requests (PRs) and issues in the project repository (https://github.com/vertical-cloud-lab/tensegrity-optimization). Citations use the reference list at the end of this document, which is numbered independently of the main text.

S1 Positioning Against Prior Work

Table S1 positions the present campaign against selected prior efforts spanning the three elements it combines: impact-tested printed tensegrity-inspired structures, rigid-flexible additive fabrication, and Bayesian optimization (BO) of architected materials. The main text (Section 2) discusses each of these studies. None of them combines a multi-material fused-deposition-modeling (FDM) tensegrity-inspired architecture with an experiment-driven optimization loop.

S2 Printability Screens and As-Built Resolution

The constant-mass projection of the main text (Section 3.1) rescales every candidate design before printing. This section records the two printability screens applied to the projected designs and what the print record shows about them, and the as-built scatter in mass.

Printability screens. We also checked each projected design against two printability limits. Both were recorded as warnings, and neither was used to reject or change a design. The first compares $\pi R^2 H$ with $250\,\mathrm{cm}^3$, where R is the radius of the circle through the joint centers and H is the height between joint centers. The $250\,\mathrm{cm}^3$ value is a placeholder size limit carried over from an earlier simulation study. It is not the printer's build volume, and because it is measured to the joint centers it undercounts the printed part. Seven printed articles (five designs) exceeded it. The second limit is a 3.0 mm minimum on the printed TPU cable diameter, adopted after the

Table S1: Positioning of this work against selected prior efforts spanning the elements combined here. The present campaign uses physical drop measurements of the transmissibility (t_{180}) and a mass-weighted form of the velocity loss index (E_{reb}, Eq. (1) of the main text) as its BO responses. It does not optimize specific energy absorption or crush energy.

Study	Architecture	Fabrication	Optimization	Response optimized	Ground truth
Pajunen et al. 2019 [1]	Truncated-tetrahedral tensegrity-inspired	Single-material 3D print	None (single condition)	n/a	Experiment (drop impact)
Intrigila et al. 2022 [2]	Bistable tensegrity-like lattice unit cell	Single-material 3D print	None	n/a	Experiment (quasi-static compr.)
Mo et al. 2023 [3]	Architected lattices	n/a (simulation)	Multifidelity BO	Simulated response metric	Simulation (+ sparse high-fid.)
Almeida 2025 [4]; Davami 2025 [5]	Tensegrity-inspired structures	Additive manufacturing	None reported	n/a	Experiment (high-rate/dynamic)
Wang et al. 2026 [6]	Tensegrity-inspired rigid-flexible metamat.	Integrated dual-material	None reported	n/a	Physical validation
This work	Tensegrity-inspired T_3-prism cell (PLA/TPU)	Multi-material FDM (PLA + TPU)	Experiment-driven BO (SAASBO/qNEHVI)	t_{180}; provisional E_{reb}	Physical experiment

thin, unsupported top cables of early PETG test prints failed. Because the projection rescales the whole cell, a design at the 3.0 mm lower bound of the cable variable can print thinner. Nine articles printed with cables of 2.51 to 2.78 mm, including all four batch-4 articles that reduced the peak shock (main text, Section 4.2). With hand-painted tendon supports (Section S4), all nine printed. One had stringing, one had a slight tendon indent, and seven had no logged defect. The most common tendon defect was a TPU strand that did not fuse to the tendon and came away from it. It was logged on five articles, all with cables of 3.28 to 4.23 mm. The 3.0 mm value is therefore a conservative rule for this workflow. Our prints do not show that thinner cables cause more defects.

As-built resolution. The nozzle diameter, the slicer line width, and support interaction discretize the as-built cross-sections, so the effective search granularity is coarser than the nominal continuous box. The 3.0 mm lower bound on the TPU cable diameter is likewise a limit of this workflow. It is the printability minimum described above, set by printing unsupported cables and by the slicer's auto-support detection on this H2D printer. Two sets of repeated prints bound the fabrication scatter in mass. One seed design printed three times has a mass standard deviation of 0.457 g, the value used in the observation noise model (main text, Section 3.4). The batch-3 plate, also printed three times, averaged 19.59, 19.86, and 20.07 g per print, so its second and third prints ran 0.27 g and 0.48 g heavier article for article. That is a per-print flow offset rather than random scatter.

S3 Joint-Design Study

We developed and ranked the strut-to-cable junction through a five-design OpenSCAD study (anchor-bulb, dovetail, TPU-sleeve overmold, eyelet-loop, and TPU-rebar variants). The working prototype uses a dovetail joint (Design B) with an anchor-bulb backup (Design A) for sensitivity studies. A captive TPU core is routed inside the PLA shell so that the soft tendon is protected from layer-line failure at the transition.

Full results, renders, and the Phase-3/Phase-4 design reviews are documented in PR #39 (joint design) and PR #35 (CAD parameterization and print history).

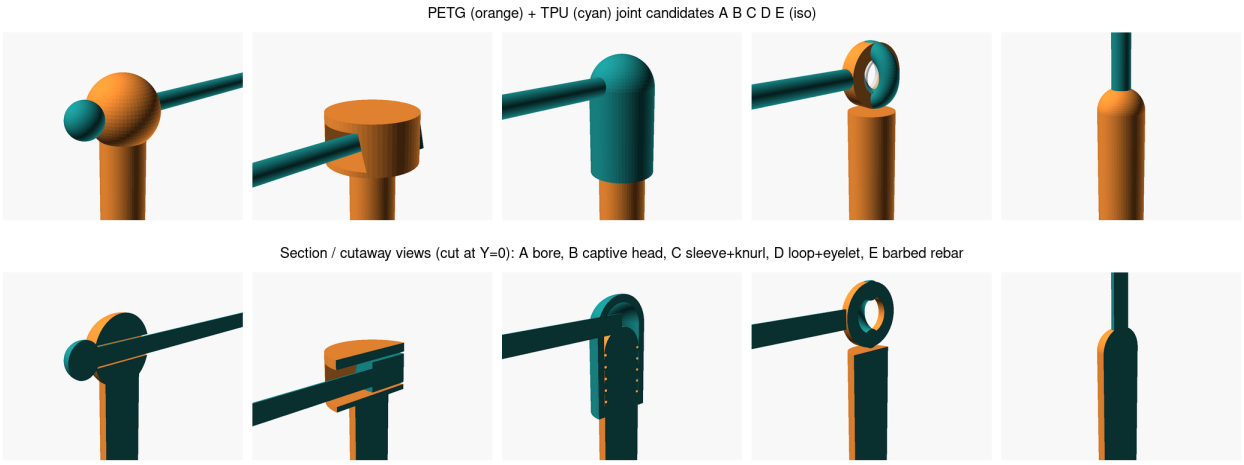


Figure S1: Five candidate strut-to-cable joint geometries from the OpenSCAD joint study (left to right: A anchor-bulb, B dovetail, C TPU-sleeve overmold, D eyelet-loop, E TPU-rebar; top row isometric, bottom row section views). The study renders use the polyethylene terephthalate glycol (PETG, orange) and TPU (cyan) material pair under which the joint family was first developed. The working prototype adapts the selected dovetail (B) with an anchor-bulb (A) backup to the PLA/TPU pair of the main text, with the captive TPU core routed inside the PLA shell.

S4 Support Strategy for Soft Members

Near-vertical TPU tendons are otherwise unsupported during printing. For the printed campaign batches, we disabled automatic support generation and painted PLA edge supports manually in Bambu Studio along each tendon. We tightened the support-to-object XY clearance from 0.35 mm to 0.2 mm during tuning on the S0 reference prism (issue #98). Support removal is the leading source of the cosmetic defects that we log per article in the print key (Section S6). They are surface stringing and, on two seed articles, a TPU strand that did not fuse to a bottom tendon and came away from it.

In parallel, we developed an automated alternative. It combines a tensegrity-specific Bambu Studio recipe (support threshold angle lowered from 40° to 10°, support material matched to the rigid extruder) with manually generated narrowing pillars that ray-cast against the printable mesh underside. The narrowing-pillar workflow is documented in PR #66, and the H2D multi-part material assignment fix in PR #64.

S5 Print Parameters and Process Control

Table S2 lists the multi-material print parameters, transcribed from the archived as-printed Bambu Studio project. The joint geometry is described in Section S3 and the support strategy in Section S4.

Moisture and nozzle management. TPU is hygroscopic, so we feed the soft filament from a heated dry box held near 8 % relative humidity, logged per print. Midway through the seed batch the TPU extruder developed a progressive drop in flow (a partial clog, compounded by a stray PLA fragment found inside the extruder). We resolved it by performing a cold pull and moving the TPU to a dedicated high-flow 0.6 mm nozzle, after which print quality was restored. We identify the affected and post-fix articles in the print key so that any nozzle-related covariate can be checked

against the measured objectives.

Table S2: Multi-material print parameters on the Bambu Lab H2D, transcribed from the archived as-printed Bambu Studio project (`t3-prism-bo-batch.H2D-MM-PLAstruts-TPUcables.as-printed.3mf`, Bambu Studio 2.7.1). Filaments are Bambu Lab PLA Basic (struts, support material) and Bambu Lab TPU 85A (tendons). The print profile is "0.30 mm Standard @BBL H2D 0.6 nozzle." The reporting structure follows the methods section of Ye et al. [7].

Parameter	PLA (struts)	TPU 85A (tendons)
Nozzle	0.6 mm	0.4 then 0.6 mm HF[a]
Nozzle temperature (°C)	220	220
Bed temperature, profile (°C)	55	35
Layer height / line width (mm)	0.30 / 0.62	
Infill	15% grid, 2 walls	near-solid
Outer / inner wall speed (mm/s)	200 / 300 (profile)	
Filament drying	n/a	dry box, ∼8% RH
Build-plate type	Textured PEI	
Build orientation	Upright (main text, Fig. 3)	
Supports	Manual painted tree supports, PLA	

[a]The TPU side ran a standard 0.4 mm nozzle until the mid-batch clog, then a dedicated high-flow (HF) 0.6 mm nozzle. Affected articles are flagged in the print key.

S6 As-Printed Campaign Record (Print Key)

Table S3 is the print key for the seed campaign: every article's print identifier, its design, its weighed mass, and its logged defects. The authoritative machine-readable key (`bo/t3-prism-bo-batch-print-key.csv`) and the as-printed slicer project (one specimen per named build plate) live at PR #102. Two entries deserve care. We printed design 08 three times. The plate label in the archived slicer project and a contemporaneous issue comment disagree on which article is the official one (*dea4ls* versus *bag26v*), and pending confirmation we use *bag26v* in the analysis. The triplicate prints of design 08 also supply the only print-to-print repeatability number for a seed design, a mass standard deviation of 0.457 g. Three prints of a single geometry do not give a general process repeatability estimate. Separately, one tested article (*amdjwm*) has no confirmed design mapping and is excluded from the surrogate fit until identified.

Table S4 is the corresponding logged key for the three model-recommended batches. Every batch-2 and batch-3 article printed at about 11% dry-box relative humidity and every batch-4 article at about 7%. The batch-3 plate was printed three times. The second-print articles (2dran1 to 2dran9) weighed 19.48 to 20.32 g, on average 0.27 g heavier than the matching first-print articles, and were all logged defect-free. The third-print articles (dran31 to dran39, printed 2026-09-28) weighed 19.68 to 20.56 g (mean 20.07 g), and the print log notes no defects. The lab record confirms each 2dran$N$ and dran3N as a re-print of drran$N$ at the same Ax trial, so the three prints pair article for article (Table S8). The nine second-print articles entered the surrogate fit as distinct completed trials at their own weighed masses and per-session standard errors. The third print was drop-tested on 2026-09-28, 2026-09-29, and 2026-10-02 (20 drops each, all captures clean) and has not yet entered the fit. Its records are on PR #86. The authoritative machine-readable keys, with plate positions and the mapping evidence per row, are the `t3-prism-bo-round1/round3/round4-print-key.csv` files archived with the manuscript data.

Table S3: Print key for the seed campaign (nine Sobol designs plus the S0 reference). Masses weighed on a 0.01 g scale. TPU fed from a heated dry box near 8% relative humidity where logged. The first five articles (printed before the TPU nozzle fix) used the standard 0.4 mm TPU nozzle, and the rest used the high-flow 0.6 mm nozzle.

Design	Article	Mass (g)	Logged defects
Spec 00	9hhbkp	21.62	none
Spec 01	6lhxfy	18.50	slight strut misprint; divot on one top tendon
Spec 02	autv5r	22.04	minor tendon misalignment near top
Spec 03	ebdna8	19.61	two strands separated from bottom tendons (support removal)
Spec 04	nvxsrv	20.66	misalignment ring after printer pause
Spec 05	6nheas	21.73	two strands separated from bottom tendons (support-related)
Spec 06	1zm8rv	18.50	stringing along diagonal tendons
Spec 07	ajhby6	20.53	filament line pulled from bottom struts
Spec 08	dea4ls	22.29	stringing; residual PLA on tendon
Spec 08	bag26v	21.42	some stringing on tendon
Spec 08	ghmj4y	22.10	inconsistent tendon cross-section near top
S0 ref.	bpx68c	20.23	none major

S7 Drop-Tower Protocol and Rig Characterization

The campaign fixture is a Lansmont Model 23 shock test system with a Test Partner 4 data-acquisition console. This section summarizes the development history behind the protocol in the main text.

Protocol. The drop height is 60 in, with the specimen upright on the carriage plate. We discard the first two drops of each session as seating warm-up. Session length differs by batch. Seed sessions contribute 101 valid captures per article (103 for autv5r), the batch-2 record lists 21 to 22 valid captures per article, and batch-3 and batch-4 sessions recorded 20 drops per article. Exact per-session counts live in the archived per-drop records, whose two generations differ on whether the count is quoted before or after the warm-up discard. Section 3.3 of the main text justifies the shortening. The statistical minimum from the pre-campaign variance study is five stabilized drops, and a truncation replay of the seed sessions preserves the full-session transmissibility ranking at twenty drops. The three prints of batch 3 also showed that the error budget is dominated by seating and print variation rather than session length. We record three high-speed video clips (959 fps) per specimen. From batch 4 on, we film them at drops 1, 10, and 20 so that clips pair deterministically with captures. The camera cannot resolve the millisecond-scale impact pulse (one to two frames), so the accelerometer channels produce the objectives and the video documents the post-impact rebound and recovery. Synchronized laser-Doppler vibrometry is reserved for follow-up characterization of selected designs. Testing takes roughly 42 s per drop, so about ninety minutes per seed article and fifteen per later-batch article. Capture settings: 1.25 MHz sample rate, 100 ms record, 2 ms pre-trigger, 150 G trigger on the base channel, AC-coupled ICP conditioning. After a settings-loss event, we re-verified the console settings digit-for-digit against an archived screenshot and confirmed data continuity to within about 1%. Per the review instruction of 2026-09-02, the per-session capture counts above should be re-verified by a person against the committed per-drop records before submission.

Table S4: Print key for the model-recommended batches, first prints only. The text summarizes the second and third prints of the batch-3 designs. Masses weighed on a 0.01 g scale.

Batch	Trial	Article	Mass (g)	Logged defects
2	10	r2d2c9	17.91	none
2	11	r2d2c4	19.33	string on one top tendon; partial TPU detachment
2	12	r2d2c2	17.96	none
2	13	r2d2c1	19.38	detached TPU string, two top tendons; odd foot support
2	14	r2d2c6	19.11	none
2	15	r2d2c7	18.51	none
2	16	r2d2c3	18.63	none
2	17	r2d2c8	19.98	full-span strings, two bottom tendons
2	18	r2d2c5	23.47	none
3	28	drran5	19.88	tiny bubbles, diagonal tendons
3	29	drran4	19.92	none
3	30	drran9	19.34	none
3	31	drran3	19.92	tiny bubbles, diagonal tendons
3	32	drran7	20.06	tiny bubbles, diagonal tendons
3	33	drran2	19.33	none
3	34	drran6	19.24	tiny bubbles, diagonal tendons
3	35	drran8	19.32	none
3	36	drran1	19.31	none
4	37	corny7	19.62	none
4	38	corny2	20.21	none
4	39	corny8	19.64	none
4	40	corny5	20.58	none
4	41	corny6	19.45	none
4	42	corny1	20.20	TPU mixed into one PLA strut layer
4	43	corny9	19.46	none
4	44	corny4	20.56	small pores, diagonal tendons
4	45	corny3	20.15	none

Impact surface. A blinded 3×3 crossover experiment over three candidate polyurethane mat arrangements (single 0.25 in Shore 70 OO, single 0.5 in Shore 60 OO, stacked) selected the single 0.5 in Shore 60 OO sheet (issue #88, PR #86). Unlike the earlier felt stack, the mat does not measurably compact over hundred-drop sessions. Sessions recorded as "5 felts" in early protocol documents were physically four felt sheets plus one cardboard layer.

Sensors and calibration. Base input: single-axis piezoelectric accelerometer on the carriage plate (1.059 mV/G, full scale 9443 G). Top response: Dytran 3133A4 triaxial accelerometer (X/Y/Z 0.690/0.667/0.734 mV/G) seated in a printed pocket at the top vertex, mounted per ISO 5348. We retired an earlier single-axis channel after range clipping. We cross-calibrated the two base-capable channels in place over fifteen co-located drops. The ratio was 0.953 ± 0.001 at CFC-180 (per-drop 0.952 ± 0.004) and stable across drop heights (PR #74).

Filtering correction. We implemented the SAE J211/1 channel-frequency-class filter used for all reported numbers from the standard's published coefficients. An earlier in-house implementation passed the two-pass (forward-backward) corner frequency to a single-pass design, which narrows every channel class by roughly 20% (an effective CFC-144 where CFC-180 was intended). We identified the discrepancy during the analysis audit of issue #94 and corrected it before the campaign numbers were produced.

Processing pipeline. For each capture, we calibrate and CFC-filter every channel independently. We form the top-vertex magnitude $\|\mathbf{a}_{\text{top}}(t)\|$ from the three filtered triaxial components and search for the two peaks entering t_{180} within a short window (approximately 1.5 ms) of first contact, so the peaks need not be simultaneous. We estimate the baseline from the record tail, because at healthy tower speed the mat-contact foot begins before the 2 ms pre-trigger window. Δv is the integral of the baselined base channel over the impact pulse and doubles as the rig-health check. Drops are discarded per the warm-up rule. Table S5 collects the operational analysis parameters in one place. Values known from the protocol are stated numerically. The remaining thresholds are to be transcribed verbatim from the version-tagged campaign analysis code at its archival release, so that the table matches the code of record.

Table S5: Operational analysis parameters for the drop-tower pipeline. "Code of record" marks values to be transcribed from the version-tagged analysis release.

Parameter	Value
Drop height	60 in (1.524 m)
Sample rate / record / pre-trigger	1.25 MHz / 100 ms / 2 ms
Trigger	150 G on the base channel
Reporting filter class	CFC-180 (SAE J211/1), two-pass corner corrected
Diagnostic filter class	CFC-1000
Peak-search window	≈1.5 ms from first contact
Baseline estimate	record tail
Warm-up exclusion	first 2 drops per session
Captures per session	101 seed (103 autv5r); 21-22 batch 2; 20 batches 3-4
Rig-health check	Δv from baselined base channel
Ringdown fit gate	$r^2 \geq 0.85$ (exponential fit)
Second-contact threshold and window	code of record
Per-capture exclusion rules	code of record
Stabilized-mean convention	code of record
Filter initialization / boundary handling	code of record

Rig health and repair history. We check rig health against the impact velocity change Δv recovered from the base channel. Over an uninterrupted 100-drop session the Δv coefficient of variation (CV, the standard deviation divided by the mean) was 0.54% with a flat drift slope. Within a session, Δv and the CFC-180 input typically ease down by 1 to 4% as the polyurethane mat warms, and recover fully after a rest of minutes. After a mat rest or re-seat, the first 5 to 10 drops carry the settling transient, which is why the warm-up discard exists. The arrest velocity has visibly stabilized as cumulative drop count grew, so we keep the Δv record. During commissioning, two guide-rod locking pins failed, and we discovered and recommissioned a defeated pin-park safety interlock. The pin-related friction produced a measurable Δv deficit that resolved after repair (issue #92). We have performed more than 4,400 instrumented drops across roughly

80 printed specimens over about four months of commissioning, characterization, and campaign sessions (tallied through issue #101 plus the batch records). The campaign batches themselves contribute about 1,500 stabilized captures including the second print of batch 3, and its third print adds 162 more.

Seating sensitivity, the between-seat rule, and the seat gauge. The three prints of batch 3 (main text, Section 4.2) showed that a session can be internally stable and still misrepresent its design. The 1.251 session passed every within-session check, and the same design's two later prints read 1.029 and 1.045. Two standing conventions followed. First, under the *between-seat replication rule*, any single-seating value that would drive a decision requires confirmation on an independent re-seat. Second, we defined an advisory *seat gauge*, the broadband ratio of CFC-1000 to CFC-180 transmissibility. It reads 1.00 to 1.07 on most sessions, and we treat a session reading 1.15 or more as a re-seat candidate. The third print is the first replicate data collected after the gauge was defined (2026-09-12). Its one gross outlier (dran35, 10.1% above its design's median over the three prints) read 2.43, and its other eight sessions read 1.01 to 1.06. Over all 27 batch-3 sessions, the four reading 1.15 or more were each their design's highest value, and the gauge correlates with the deviation from the design median (Spearman $\rho = 0.52$, $p = 0.006$). Both conventions, with the session-by-session drift-watch record, live in the drop-test datasets on PR #86. The three-print comparison is archived with the manuscript data (`t3-prism-bo-round3-three-print.csv`, generated by `scripts/figures/round3_three_print_analysis.py`).

Objective provenance and adversarial review. The objective stack evolved through a measurability review (PR #97). Per-specimen crush energy in joules is not recoverable from this rig, because it has no falling-mass or displacement channel and the specimens stay elastic. The campaign responses are therefore the filtered peak-acceleration ratio and a provisional velocity loss index, optimized in a mass-weighted form, with quasi-static crush metrics deferred to a load-frame campaign. We pre-registered an adversarial review of the objective choices with an external AI referee (Edison ANALYSIS task `3e398131`, archived in the repository). It recommended accounting for article-level rather than per-drop noise, treating the velocity loss index as a candidate constraint pending a restrained-versus-unrestrained validation with synchronized video, and reserving print capacity for replicate articles. The main text records how the campaign responded to these recommendations. We printed and tested the batch-3 plate three times, which measured the print-to-print scatter directly and produced the between-seat replication rule. The velocity loss index remains a candidate constraint, and the surrogate continues to ingest per-session standard errors, with the replication evidence carried as an interpretation bound rather than as an inflated noise term. We plan to act on the reserved-replicate recommendation through the replicate rank-stability study of the main-text Methods. That study will re-print the best-, mid-, and worst-predicted in-sample designs in nominal triplicate and test whether the predicted ordering survives print-to-print and seating variation.

S8 Planned Quasi-Static Extension

The drop test loads the specimens in their elastic regime. The crush lasts 1 to 3 ms and the article recovers, so crush-energy metrics such as specific energy absorption (SEA) and compaction efficiency are not recoverable from these captures by any reprocessing. We plan a quasi-static compression campaign on a benchtop load frame (BYU CB 123 Instron, with a low-range load cell

appropriate to the expected cell stiffness) to supply the complementary crush metrics

$$\mathrm{SEA} \;=\; \frac{1}{m}\int_0^{\delta_d} F(\delta)\,\mathrm{d}\delta, \qquad \eta_c \;=\; \frac{\int_0^{\delta_d} F(\delta)\,\mathrm{d}\delta}{F_{\max}\,\delta_d}, \tag{1}$$

with the densification displacement δ_d taken at the maximum of the energy-absorption efficiency curve and $F_{\max}$ the peak force within $[0, \delta_d]$. We follow the standard cellular-solid treatments [8–11] and mirror the quasi-static protocol of Pajunen et al. [1]. These metrics will be reported as a complementary characterization of selected designs rather than as objectives of the drop-tower campaign. Slip resistance and traction at the specimen-to-floor interface, governed by the tip geometry rather than the internal architecture, follow ISO 11334-4 [12] for walking-aid tips and are out of scope for the present axial-impact study.

S9 Printed-Mass Model

We projected the seed batch onto constant *solid* CAD mass ($m^\star = 30.95\,\mathrm{g}$), yet the articles weigh 18.50 to 22.29 g. The gap arises because PLA struts print at partial infill (15% grid, two walls) while thin TPU cables print near solid, and the PLA-to-TPU volume split swings widely across the design box. A two-stage calibrated model closes this gap (PR #102, `bo/t3_prism_mass_model.py`). Analytic member volumes are corrected to rendered-solid grams (one factor per material). Rendered-solid grams are then mapped to weighed grams through a wall-plus-infill term in the as-printed strut diameter (fitted infill 23.6%, wall 1.18 mm, TPU factor 0.744). The model's residual standard deviation is 0.378 g. For context, the three available prints of design 08 have a sample standard deviation of 0.457 g. Three prints of one geometry are insufficient to establish a general process repeatability limit. Run in reverse inside the projection, the model holds *printed* grams constant without running the slicer. For a target mass, the projection finds the uniform scale factor at which the predicted printed mass equals the target, and we print the design at that scale. Bisection is enough for this, because predicted mass rises monotonically with scale. Batches 3 and 4 used a target of 20.23 g, the mass of the printed reference article. Batch mass CV fell from 5.5% (tested seed articles) and 8.7% (batch 2, still solid-mass-projected) to 1.7% (batch 3) and 2.3% (batch 4). The weighings sit at or below the model's estimate as a per-print flow offset (-0.52, -0.25, and -0.04 g for the three prints of batch 3 and -0.26 g for batch 4). In a first test of the model's infill terms, regressing the print-averaged mass residuals of the eighteen batch-3 articles on the two commanded infills gives slopes indistinguishable from zero over the realized 13 to 34% spread. We will recalibrate the session term with the accumulated weighings before the next batch's projection. Two batch-2 recommendation sets exist in the record. The batch actually sliced and sent to the printer was generated before this correction, under the seed batch's solid-mass projection. A subsequently regenerated set applies the constant-printed-mass projection and seeded the batch-3 tooling. Both are archived with the manuscript data (`round2-as-printed-solid-mass-projection.csv`, `round3-candidate-constant-printed-mass.csv`).

S10 Replicate Rank-Stability Study Plan

The main text (Section 3.5) describes the planned replicate study: from the campaign's in-sample designs, ranked by the current posterior's predictions, the best-predicted, the worst-predicted, and one mid-pack ("run-of-the-mill") design will each be re-printed as a nominal three fresh articles and drop-tested under the standard protocol, including the between-seat replication rule. The repeat count is provisional and will be fixed when the print is scheduled. With three designs the exact

permutation floor for a Spearman rank correlation is $p = 1/6$, so no significance claim is available at this size. Instead, the primary readout is the separation of the three designs' repeat clusters relative to the within-design (print-to-print plus seat-to-seat) scatter, reported alongside ρ_s between the predicted and measured orderings of each repeat. Clean separation would support treating the campaign's rankings as design properties. Overlapping clusters would measure the ranking noise floor that the three batch-3 prints could only bound. Table S6 gives the reporting format. We will add entries and design IDs once the repeat articles are printed and tested.

Table S6: *Placeholder (data pending).* Planned replicate rank-stability study: the best-, mid-, and worst-predicted in-sample designs, each re-printed as a nominal three fresh articles and drop-tested under the standard protocol. Entries will record each repeat's stabilized t_{180}, the within-design scatter, and the measured rank of each design's repeat cluster against its predicted rank.

Tier	Design	Pred. t_{180}	t_{180} (repeats 1 / 2 / 3)
Best	*TBD*	-	- - -
Mid	*TBD*	-	- - -
Worst	*TBD*	-	- - -

S11 Metal-Analog Validation Plan

A planned validation campaign will rebuild a subset of the T_3-prism designs as metal analogs, using the same R, H, θ, and member-diameter parameterization as the printed PLA/TPU specimens. The analogs will use hollow aluminum rods for the compression members and stainless-steel threaded cables for the tension members. To probe transfer across material systems rather than only confirming the optimum, the selected designs will deliberately span the predicted performance range. Representatives the surrogate predicts to be worst-, mid-, and best-performing will each be built in both material systems and characterized under the identical drop protocol of Section S7. We treat this as a planned, exploratory comparison rather than a preregistered validation. The deformation modes, joint compliance, friction, pretension behavior, and strain-rate sensitivity all differ between the PLA/TPU and aluminum/steel systems, so agreement is not expected a priori. The protocol details that would let the comparison distinguish geometric rank transfer from material-system and mass confounds remain to be fixed and will be tabulated before any metal article is built. They are the aluminum alloy and temper, tube wall thickness, cable construction, joint hardware and torque, article and payload masses, replicate count, and tie-handling rules. The intended rank statistic is the Spearman rank-correlation coefficient ρ_s between the two material systems' orderings of the primary campaign metric (t_{180} under the identical drop protocol), reported with an exact permutation test. Table S7 and Fig. S2 will report the measured PLA/TPU ranking and the hollow-aluminum/stainless-steel ranking on t_{180}, with the velocity loss index and, later, quasi-static specific energy absorption (SEA, Section S8) as secondary rankings. Agreement between the two primary rankings would be preliminary evidence consistent with transfer beyond the as-printed polymers. Disagreement would not isolate which material or joint difference caused the change.

Table S7: *Placeholder (data pending).* Planned rank-ordering validation: representative T_3-prism designs predicted to be worst-, mid-, and best-performing, each fabricated as a printed PLA/TPU specimen and as a hollow-aluminum-rod / stainless-steel-cable metal analog. Entries will record the measured performance metric (t_{180} under the identical drop protocol, with the velocity loss index secondary) and the resulting rank within each material system.

Predicted tier	Design ID	PLA/TPU (rank)	Al/SS (rank)
Worst	*D-w1*	-	-
Worst	*D-w2*	-	-
Mediocre	*D-m1*	-	-
Mediocre	*D-m2*	-	-
Best	*D-b1*	-	-
Best	*D-b2*	-	-

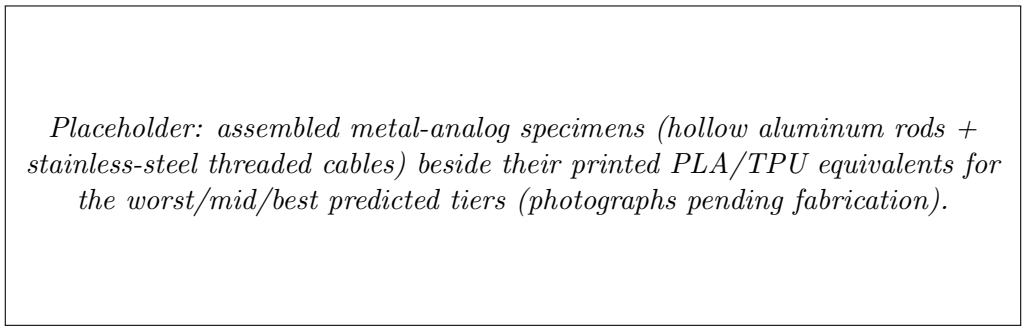


Figure S2: Metal-analog validation specimens alongside their printed PLA/TPU equivalents for each predicted performance tier.

S12 Batch-3 Three-Print Repeatability Table

Table S8 tabulates the stabilized transmissibility of every batch-3 article over the three prints of the plate, the data behind the three-print figure and the replication findings of the main text (Section 4.2). The seat gauge and the drift flag used to screen the sessions are defined in Section S7.

S13 Surrogate Diagnostic Figures

Figures S3 and S4 are the surrogate diagnostics discussed in the main text (Section 4.3): the seed-state length-scale diagnostics of the sparse axis-aligned subspace model (SAASBO) and the leave-one-design-out cross-validation of the surrogate over the full 44-article record. The cross-validation runner, its fold-level records, and the reliability-ceiling analysis of the reprint pairs are archived with the campaign code and the manuscript data (`t3-prism-bo-round5-logocv.csv`, `scripts/figures/reprint_reliability_ceiling.py`).

Table S8: Batch-3 repeatability as stabilized t_{180} over three prints of the same nine designs, under the protocol of the main-text results tables (CFC-180, stabilized means). Articles drran$N$, 2dran$N$, and dran3N are prints 1, 2, and 3. "Pred." is the posterior-mean t_{180} archived before any batch-3 article was tested. "Mean" averages the prints that pass the health screen, a seat gauge (the ratio of CFC-1000 to CFC-180 transmissibility) below 1.15 and no drift flag. § marks a failed seat gauge and † a transmissibility-drift flag. E_{reb} is listed for prints 1, 2, and 3.

Trial	N	Pred.	Print 1	Print 2	Print 3	Mean	E_{reb} (mJ)
28	5	1.059	1.0373	1.0604	1.1679§	1.0489	5.0 / 5.0 / 7.2
29	4	1.032	1.0344	1.0433	1.0201	1.0326	9.1 / 11.2 / 15.5
30	9	1.058	1.0425	1.0451	1.0438	1.0438	5.2 / 15.1 / 6.7
31	3	1.043	1.0389	1.0533	1.0366	1.0429	4.2 / 9.4 / 7.0
32	7	1.020	1.2513§	1.0286	1.0448	1.0367	8.2 / 11.6 / 10.8
33	2	1.056	1.0345	1.0900§	1.0386	1.0366	11.8 / 4.5 / 4.5
34	6	1.065	1.0471	1.0859†	1.0353	1.0412	4.8 / 7.7 / 6.8
35	8	1.039	0.9804†	1.0109	0.9958	1.0033	11.2 / 9.9 / 8.3
36	1	1.051	1.0326	1.0791§	1.0209	1.0267	18.1 / 7.3 / 9.6

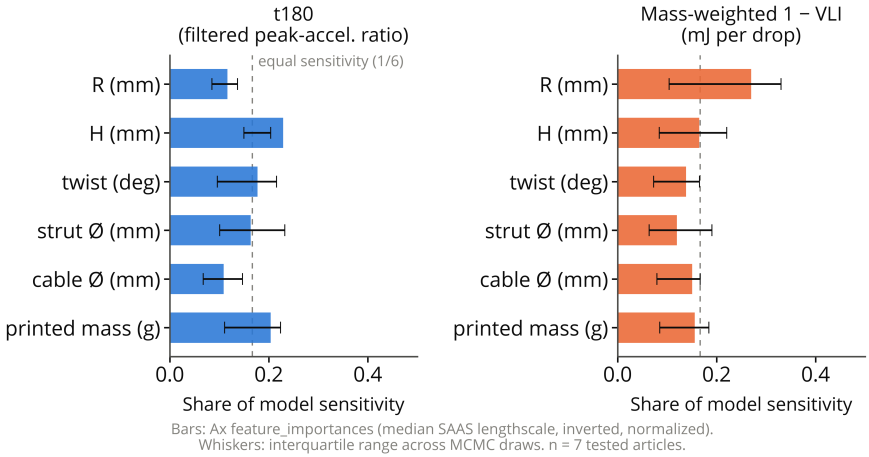


Figure S3: Seed-state SAASBO length-scale diagnostics per objective for the five design variables plus printed mass. Bars are median normalized inverse length scales over MCMC draws, and whiskers are interquartile ranges. The horizontal axis, labeled "share of model sensitivity" in the plot, is this normalized inverse-length-scale diagnostic. No input separates decisively from the equal-share line (dashed) at seven mapped articles, and these prior-sensitive model diagnostics are not evidence of variable relevance.

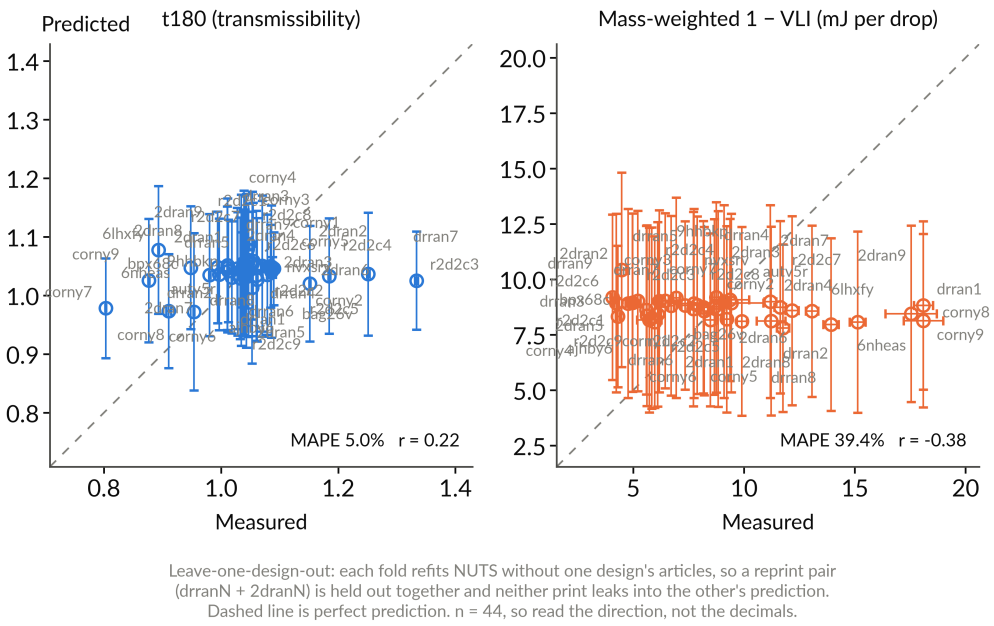


Figure S4: Leave-one-design-out cross-validation of the SAASBO surrogate over the full 44-article record (35 design folds, one full NUTS refit per fold). Each batch-3 reprint pair is held out together so neither print leaks into the other's prediction. Vertical bars are one posterior standard deviation of the held-out prediction, horizontal bars are the ingested standard error, and the dashed line is parity. Held-out t_{180} predictions stay close to the pooled mean (MAPE 5.0%, rank correlation $+0.08$), and the E_{reb} ordering is anti-correlated (-0.38). Design-level held-out skill is near zero on transmissibility and absent on the mass-weighted velocity-loss objective.

References

[1] Kirsti Pajunen, Paul Johanns, Raj Kumar Pal, Julian J. Rimoli, and Chiara Daraio. Design and impact response of 3D-printable tensegrity-inspired structures. *Materials & Design*, 182: 107966, 2019. doi: 10.1016/j.matdes.2019.107966.

[2] Claudio Intrigila, Andrea Micheletti, Nicola A. Nodargi, Edoardo Artioli, and Paolo Bisegna. Fabrication and experimental characterisation of a bistable tensegrity-like unit for lattice metamaterials. *Additive Manufacturing*, 57:102946, Sep 2022. ISSN 2214-8604. doi: 10.1016/j.addma.2022.102946. URL https://doi.org/10.1016/j.addma.2022.102946.

[3] Changyu Mo, Paris Perdikaris, and Jordan R. Raney. Accelerated design of architected materials with multifidelity Bayesian optimization. *Journal of Engineering Mechanics*, 149(6): 04023028, 2023. doi: 10.1061/JENMDT.EMENG-7033.

[4] Nara Almeida, Ben Gulledge, Jesse Park, Kasra Momeni, Ali Tajyar, Keivan Davami, and Sadie Beck. High strain rate response of 3d-printable tensegrity-inspired structures. *International Journal of Solids and Structures*, 322:113590, 2025. doi: 10.1016/j.ijsolstr.2025.113590.

[5] Keivan Davami, Russell Rowe, Ben Gulledge, Jesse Park, Ali Beheshti, Anthony Palazotto, Fariborz Tavangarian, and Sadie Beck. Dynamic analysis of additively manufactured tensegrity structures. *International Journal of Impact Engineering*, 198:105208, 2025. doi: 10.1016/j.ijimpeng.2024.105208.

[6] Zhenyu Wang, Meiling Zhang, Jianwei Sun, Qi Wen, and Jinkui Chu. Integrated fabrication and validation of tensegrity-inspired rigid-flexible mechanical metamaterials. *Additive Manufacturing*, 118:105107, 2026. doi: 10.1016/j.addma.2026.105107.

[7] Haitao Ye, Qingjiang Liu, Jianxiang Cheng, Honggeng Li, Bingcong Jian, Rong Wang, Zechu Sun, Yang Lu, and Qi Ge. Multimaterial 3D printed self-locking thick-panel origami metamaterials. *Nature Communications*, 14:1607, 2023. doi: 10.1038/s41467-023-37343-w.

[8] Lorna J. Gibson and Michael F. Ashby. *Cellular Solids: Structure and Properties*. Cambridge Solid State Science Series. Cambridge University Press, 2nd edition, 1997. doi: 10.1017/CBO9781139878326.

[9] M. Avalle, G. Belingardi, and R. Montanini. Characterization of polymeric structural foams under compressive impact loading by means of energy-absorption diagram. *International Journal of Impact Engineering*, 25(5):455–472, 2001. doi: 10.1016/S0734-743X(00)00060-9.

[10] P. J. Tan, S. R. Reid, J. J. Harrigan, Z. Zou, and S. Li. Dynamic compressive strength properties of aluminium foams. Part I—experimental data and observations. *Journal of the Mechanics and Physics of Solids*, 53(10):2174–2205, 2005. doi: 10.1016/j.jmps.2005.05.007.

[11] N. Michailidis, F. Stergioudi, A. Tsouknidas, and E. Pavlidou. Establishment of process parameter optimization for the manufacturing of aluminium foams and the determination of their mechanical properties. *Materials Science and Engineering: B*, 176(1):19–25, 2011. doi: 10.1016/j.mseb.2010.08.006.

[12] International Organization for Standardization. ISO 11334-4:1999 Assistive products for walking manipulated by one arm — Requirements and test methods — Part 4: Walking sticks with three or more legs. Standard, International Organization for Standardization, 1999. URL https://www.iso.org/standard/27298.html.